

Noise Study Check GUIDE

Noise Study Check

- 1-Complete Ducting Layout.
- 2- Mentioned Flowrates.
- 3- Architecture Layout
- 4-Double Line ducting Layout
- 5-Sound Power Levels.
- 6-Design NC requirement (Optional).
- 7-Maximum Allowable Pressure Drop (Optional).

Ducting Layout Check

ALL the Floors Should be attached to insure that the Ducting Layout is complete.

*If Only 1 DWG file is attached we must check that the Ducting Layout is Complete

(If this floor ducting layout looks like one of these below figures (Fig.1,2) so this ducting layout is incomplete & cann't request Noise Study.

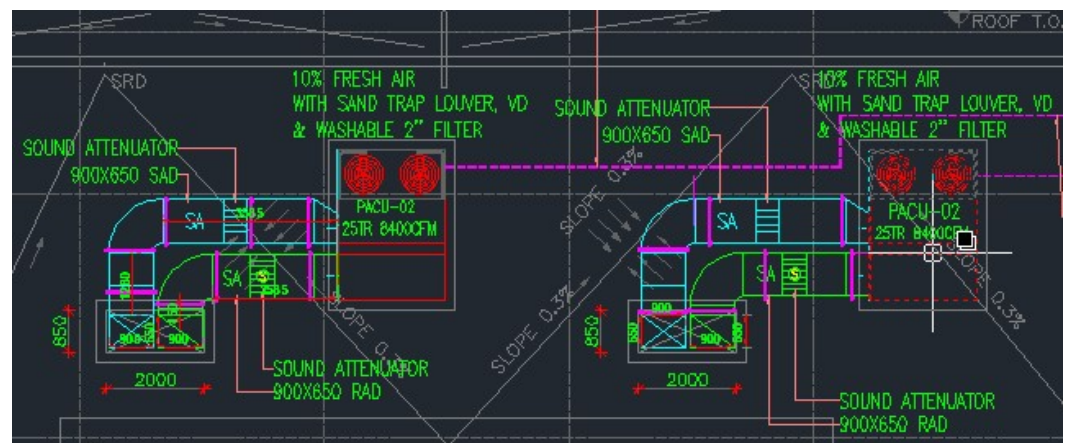


Fig.1 – Incomplete Ducting Layout (missing floors)

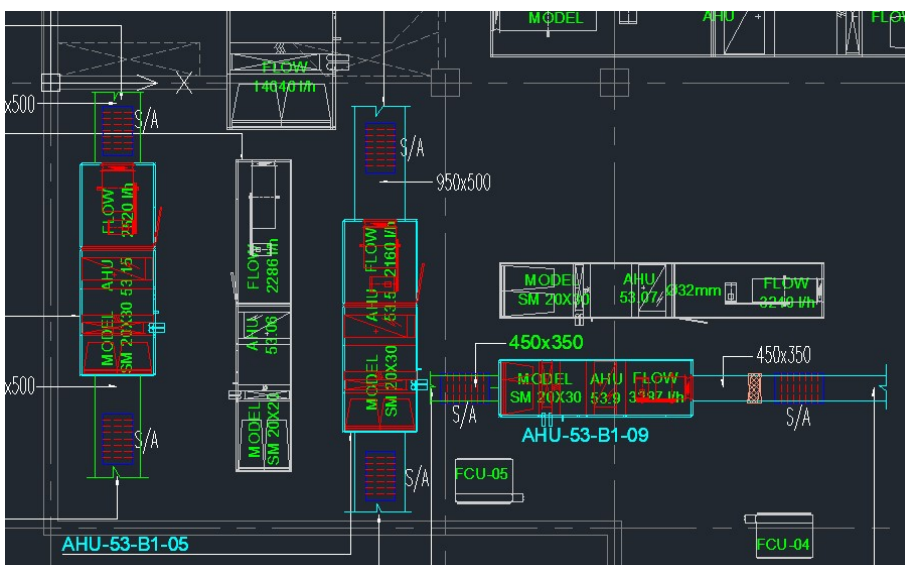


Fig.2 – Incomplete Ducting Layout

Architecture Layout Check

-The Architecture layout is important in determining the locations where the AHUs supply Air.

-It's also Important to Calculate the area at which the AHU work.

That's why we have to check for the availability of the Architecture Layout.

Fig.3 Shows an Example for Missing Architecture Layout so, If the DWGs received from customer looks like this Figure (**Fig.3**) Kindly Request a DWG with complete Architecture Layout from the Customer before Requesting the Noise Study.

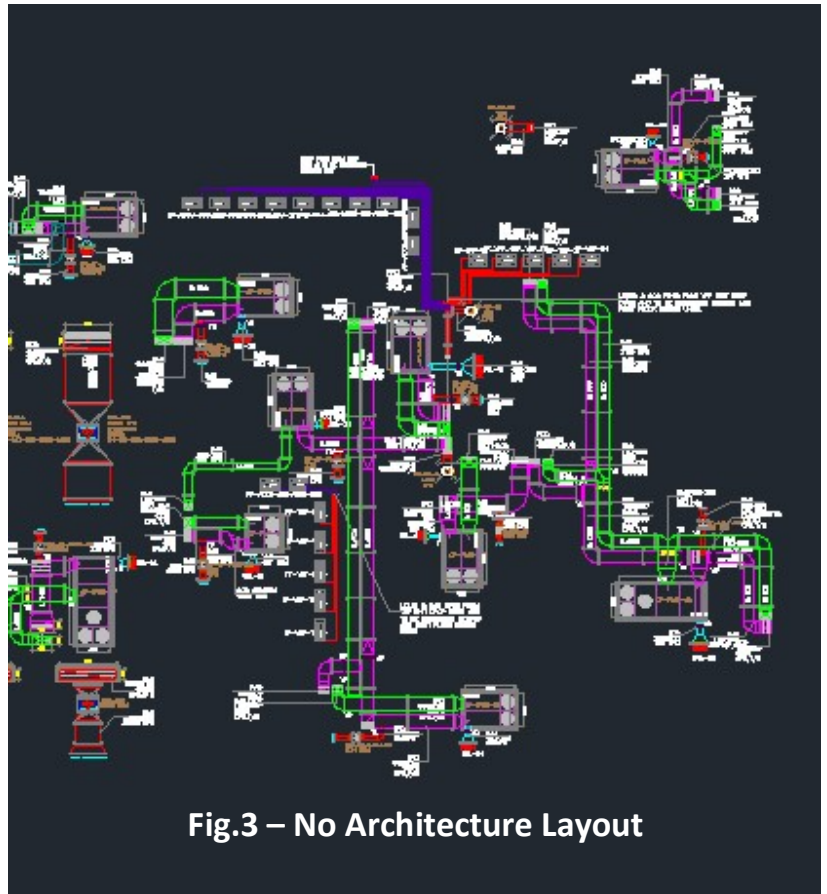


Fig.3 – No Architecture Layout

Double Line ducting Layout Check

-We Have to check that the DWGs received From Customer are in the double line form.

-The double Line Layout is important to determine the ducting sizes, places of take-off, Change in Size (Enlargement, Reducing).

-All of this information are necessary & can't be determined in case of single line Ducting layout.

-**Fig.4** Shows Single Line Ducting Layout, if the DWGs received from customer looks Like **Fig.4**, Kindly Contact the customer & request a double line ducting layout before requesting the Noise study.

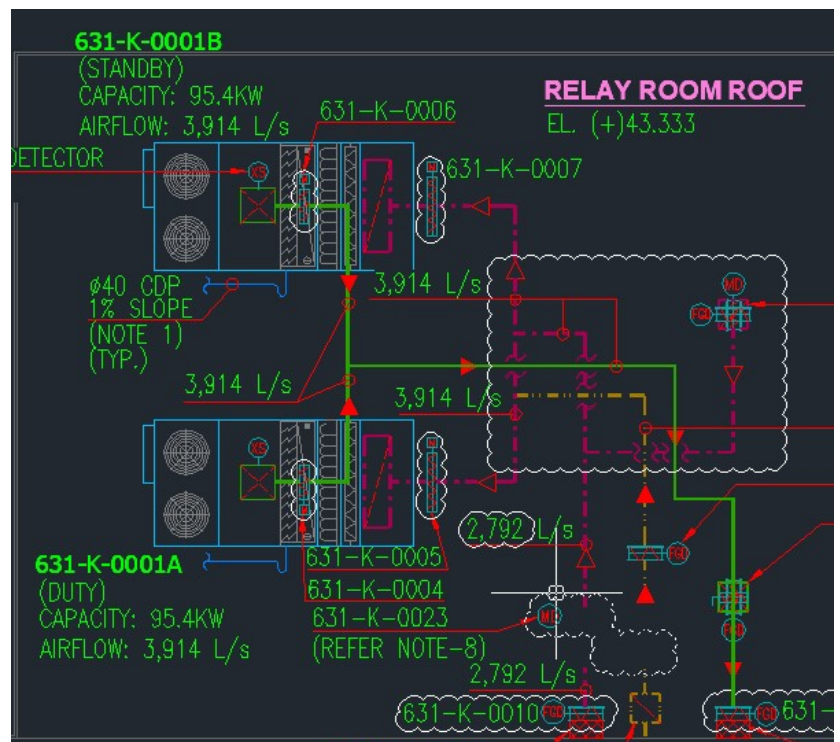


Fig.4 – Single Line ducting Layout

Sound Power Levels Check

Usually the sound Power level come in one of these Forms:

1. Sound Power levels at each Frequency.
2. Sound Pressure levels at each Frequency.
3. Sound Charts / Tables [Depend on Operating Conditions].

1. Sound Power levels at each Frequency. [\(Acceptable\)](#)

if the sound Power levels are given at each frequency like (Fig.5,6),it's acceptable .

Nameplate Model: TBI-FS-5-L42-250-HTUL

Sound Power by Octave Band

Sound Data	62.5	125	250	500	1000	2000	4000	8000	LwA	dBA
Inlet	103	112	108	106	100	95	90	86	107	96

*FLA - based on tables 150 or 148 of National Electrical Code 2002. Actual motor FLA may vary, for sizing thermal overload, consult factory.

LwA - A weighted sound power level, based on ANSI S1.4

dBA - A weighted sound pressure level, based on 11.5 dB attenuation per octave band at 1.52 m- dBA levels are not licensed by AMC

Fig.5

Packaged Unit

1.1.132.0

General Info

Customer: KADEN INVESTMENT

Project: AL SHAKREEN MOSQUE

Ref/Tag: PACU-01R

Revision: Rev 0

Done By: SIBGHAT

Location: RIYADH

Quantity: 1

Date: Wednesday, November 9, 2022

Output Performance

Selected Unit Model

PY300

Sound Data Of Evaporator Fan

Sound Power Data	Frequency (Hz)	63 (Hz)	125 (Hz)	250 (Hz)	500 (Hz)	1000 (Hz)	2000 (Hz)	4000 (Hz)	8000 (Hz)	Sum (dbA)
Outlet (db)		89.5	90.7	82.1	90.9	80.4	78.2	75.8	70.4	88.7
Inlet (db)		84.4	85.4	80.4	83.7	77.7	76.7	74.7	69.7	83.8

Sound Pressure at 1m

Sound Power Data	Frequency (Hz)	63 (Hz)	125 (Hz)	250 (Hz)	500 (Hz)	1000 (Hz)	2000 (Hz)	4000 (Hz)	8000 (Hz)	Sum (dbA)
Outlet (db)		81.5	82.7	74.1	83.0	72.4	70.2	67.8	62.5	80.8
Inlet (db)		76.4	77.4	72.4	75.7	69.7	68.7	66.7	61.7	75.9

Sound Pressure at 2m

Sound Power Data	Frequency (Hz)	63 (Hz)	125 (Hz)	250 (Hz)	500 (Hz)	1000 (Hz)	2000 (Hz)	4000 (Hz)	8000 (Hz)	Sum (dbA)
Outlet (db)		75.5	76.7	68.1	76.9	66.4	64.2	61.8	56.4	74.7
Inlet (db)		70.4	71.4	66.4	69.7	63.7	62.7	60.7	55.7	69.8

Sound Pressure at 4m

Sound Power Data	Frequency (Hz)	63 (Hz)	125 (Hz)	250 (Hz)	500 (Hz)	1000 (Hz)	2000 (Hz)	4000 (Hz)	8000 (Hz)	Sum (dbA)
Outlet (db)		69.5	70.7	62.1	70.9	60.4	58.2	55.8	50.4	68.7
Inlet (db)		64.4	65.4	60.4	63.6	57.6	56.6	54.6	49.6	63.8

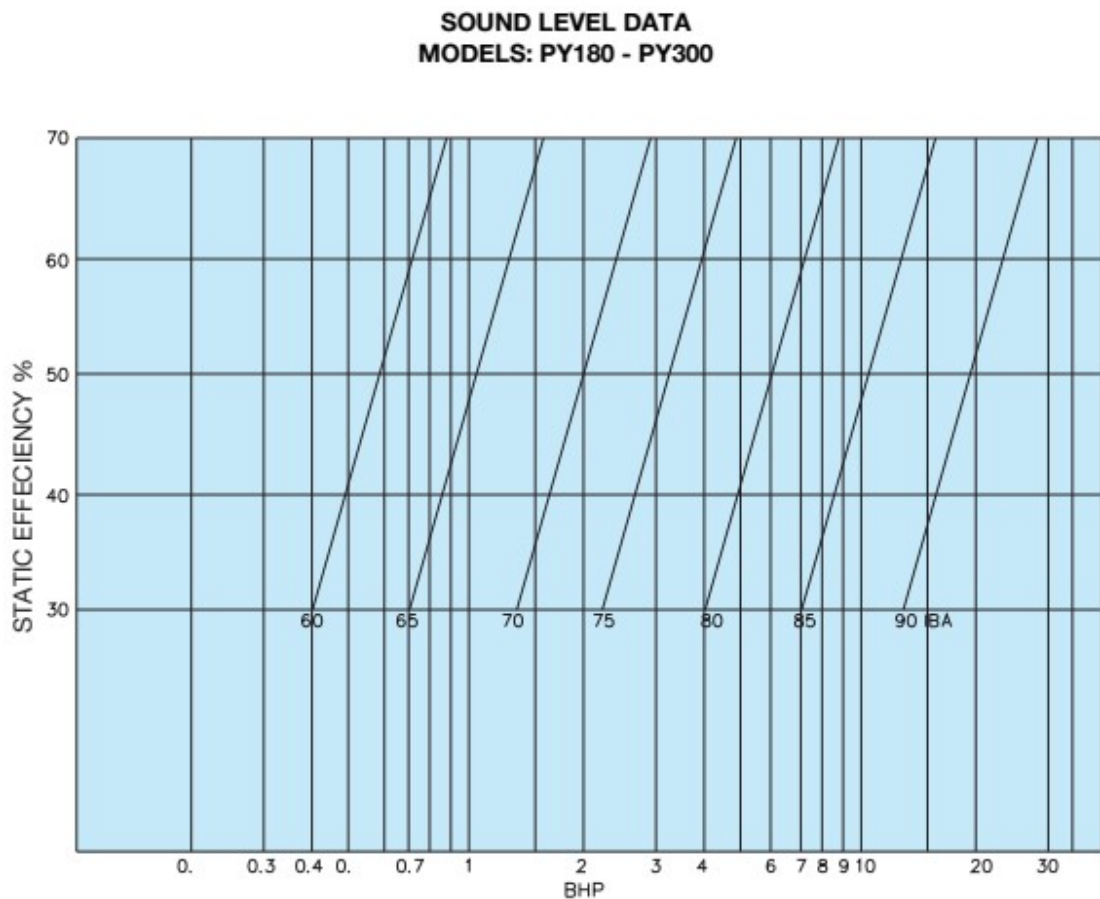
Sound Pressure at 6m

Sound Power Data	Frequency (Hz)	63 (Hz)	125 (Hz)	250 (Hz)	500 (Hz)	1000 (Hz)	2000 (Hz)	4000 (Hz)	8000 (Hz)	Sum (dbA)
Outlet (db)		65.9	67.1	58.6	67.4	56.8	54.7	52.2	46.9	65.2
Inlet (db)		60.9	61.9	56.9	60.1	54.1	53.1	51.1	46.1	60.3

Fig.6

2. **Sound Power Charts /Tables.** [\(Not-Acceptable\)](#)

-If the sound Power levels are given in the form of Charts as **(Fig.7)**, this means that the Given sound data depends on operating conditions. (Should contact the customer & ask for the Sound power levels at each wave frequency.



SOUND POWER FACTOR TABLE

VALUES GIVE THE SOUND POWER

Octave Band Center Frequency – Hz	63	125	250	500	1000	2000	4000	8000
255-443 RPM	-3.5	+4	+12.5	+8	+4.5	+1.5	-2	-7.5
443-887 RPM	-4	+3.5	+9	+10	+4	+1	-2.5	-8
887-1750 RPM	-4	+3.5	+8.5	+7	+7	+0.5	-3	-8

Fig.7 – Sound Power levels chart

-If the sound Power levels are given in the Tables as (**Fig.8**), this means that the Given sound data depends on working conditions. (Should contact the customer & ask for the working condition).

A-weighted Sound Power Level, dB(A)

See notes 1, 3 and 6-14 listed below

Configuration	Exhaust	Applied Load	Octave Band Center Frequency (Hz)											Overall Sound Power Level
			16	31.5	63	125	250	500	1000	2000	4000	8000	16000	
Standard – Unhoused	Infinite Exhaust	0% Prime	N/A	65.9	84.7	104.6	118.3	120.0	119.0	116.6	111.7	101.8	89.7	124.9
		75% Prime	N/A	66.7	85.2	105.0	118.2	119.9	118.5	115.8	111.0	105.6	93.0	124.7
		100% Prime	N/A	67.1	85.6	105.7	118.2	120.0	118.7	116.5	111.8	108.9	94.7	124.9
		100% Standby	N/A	67.0	85.8	105.6	118.1	120.0	118.8	117.2	112.5	110.5	96.3	125.1
Housed	Mounted Muffler	0% Prime	N/A	79.2	89.2	100.6	107.2	106.6	101.5	100.7	92.6	83.4	69.9	111.4
		75% Prime	N/A	77.1	98.3	102.4	107.0	106.1	101.0	99.8	92.0	95.5	71.7	111.5
		100% Prime	N/A	76.7	99.7	102.9	107.0	106.1	101.1	100.3	93.0	97.8	74.7	111.8
		100% Standby	N/A	76.5	100.8	103.3	107.1	106.4	101.5	100.9	93.9	97.9	76.0	112.1
Remote Cooling - Unhoused	Infinite Exhaust	0% Prime	N/A	63.9	82.8	100.0	105.2	113.0	115.4	114.8	109.9	99.7	87.0	120.0
		75% Prime	N/A	64.9	82.9	100.5	105.2	112.9	115.2	114.6	109.9	104.9	92.3	120.0
		100% Prime	N/A	65.2	83.6	101.2	106.0	113.0	115.6	115.5	111.1	107.8	94.4	120.7

Fig.8 – Sound Power levels Table

3. Sound Pressure levels at each Frequency.

if the sound levels are given as Sound Pressure levels as Fig.9, Should ask the customer about the **working distance**.

MRCT-300CWN1-Y(DH) octave band levels								
frequency (Hz)	63	125	250	500	1000	2000	4000	8000
Octave band sound pressure level (dB(A))	78	71	65	64	67	71	63	59

Fig.9 – Sound Pressure levels